

REVIEW

Re-evaluating antibody–drug conjugate linker stability: assessment, interpretation, and clinical translation

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Available online 21 April 2026

Antibody–drug conjugates (ADCs) have emerged as an important class of anticancer therapeutics. Their highly modular design offers multiple opportunities to refine their properties and improve clinical performance by tuning individual components, including the antibody, linker, and payload.

Recent ADC optimization efforts have focused on more stable linker designs, apparently based on the assumption that premature payload release, rather than ADC uptake and intracellular catabolism, is the primary driver of off-target toxicity and limited antitumor activity. However, clinical data highlight a more complex picture.

Several clinically successful ADCs employ relatively unstable linkers, including some with short systemic half-lives. In contrast, ADCs with increased linker stability have not consistently demonstrated improved clinical outcomes and, in many cases, have been associated with unexpected toxicities.

In this article, we examine common assumptions regarding ADC linker stability, highlight limitations of preclinical models in predicting human payload exposure, address appropriate methods to assess ADC stability, and discuss the clinical implications of altering ADC linker stability. By re-evaluating the role of linker stability in ADC design, this review aims to inform more rational strategies for the development of next-generation ADCs.

Key words: ADC, ADC disposition, antibody–drug conjugate, linker stability, clinical translation, pharmacokinetics

INTRODUCTION

Over the past two decades, substantial research has focused on enhancing antibody–drug conjugate (ADC) linker stability.¹ It is a widely held view that more stable linkers will improve the clinical success of ADCs, and that unwanted toxicities and insufficient efficacy originate from ADC linker instability, often referred to as ‘premature payload release.’² However, these are oversimplified views and overlook that the majority of the administered ADC dose is metabolized in normal tissue rather than at the tumor site.³ Clinical experience with otherwise similar ADCs exhibiting different linker stabilities suggests that the relationship between stability and clinical outcomes is more complex than often assumed.

Comparing clinical trials to enable conclusions about drug design is challenging and is associated with well-understood pitfalls. Relatively few ADC programs feature the same antibody and payload, enabling interpretation of the contribution of linker stability. Even among such closely related ADCs, evaluations have not been conducted in head-to-head trials, necessitating cautious cross-trial comparisons. In this article, care has been taken to focus on trends that emerge across multiple programs.

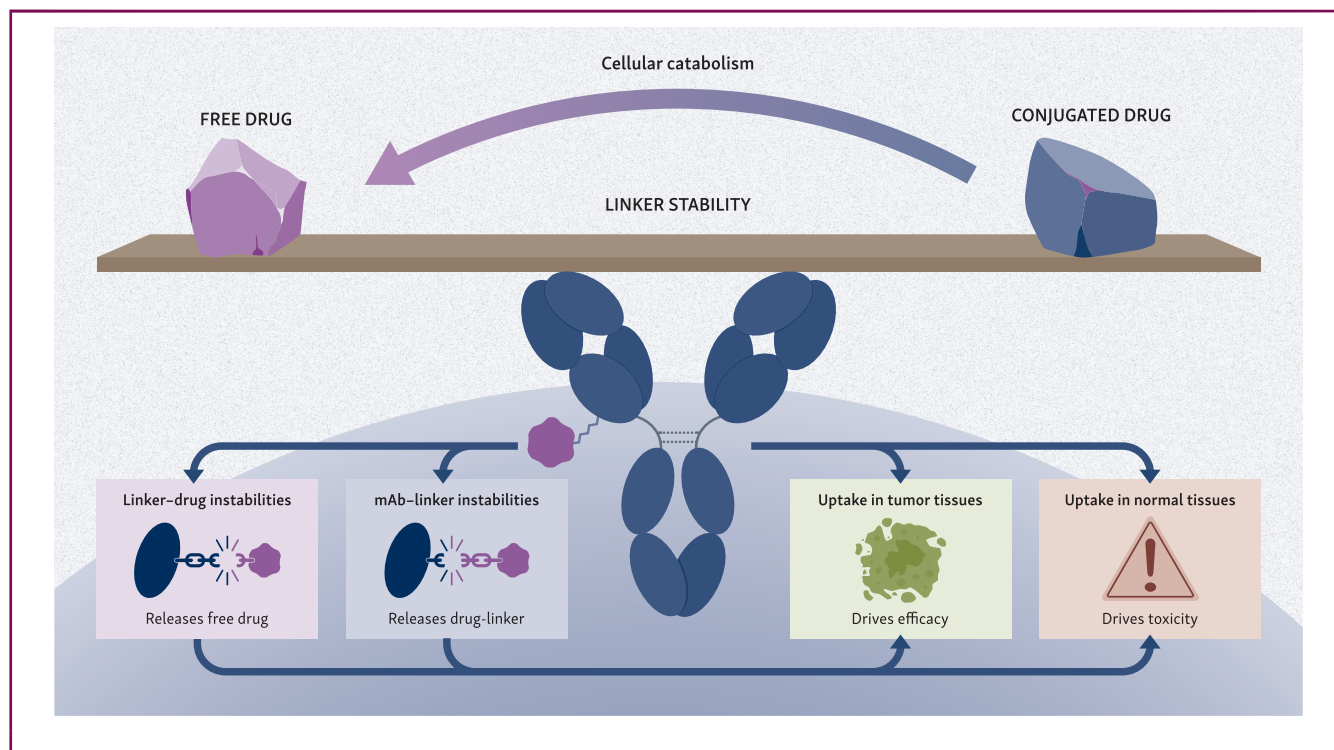
ADCs employ a broad range of linker chemistries, and their stability can be tuned through the choice of antibody conjugation site and linker design. Historically, widely used ADC linkers have shown limited stability.³ For example, hydrazone linkers used in calicheamicin and doxorubicin ADCs had half-lives of <2 days.⁴ A first-generation disulfide linker used for DM1 (SPP-DM1) also resulted in very unstable ADCs.^{5,6} Carbonate linkers, used in govitecan (CL2A-SN38) and tirumotecan (pyrimidine-CL2A-KL610023) ADCs undergo rapid hydrolysis in plasma, with half-lives of 1–2 days.^{7,8} Popular maleimide-containing maleimidocaproyl (MC) and maleimidomethyl cyclohexane-1-carboxylate (MCC) linkers (Supplementary Figure S1, available at <https://doi.org/10.1016/j.annonc.2026.04.008>), originally developed in the 1990s and employed in >200 clinical stage ADCs, commonly lose 50% of

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GRAPHICAL ABSTRACT



the initial drug-to-antibody ratio (DAR) after a week in plasma through a retro-Michael reaction.⁹ The vedotin (MC-VCit-PABC-MMAE), mafodotin (MC-MMAF), deruxtecan (MC-GGFG-DXd), and emtansine (MCC-DM1) drug-linkers are among the more widely studied examples of these MC-based ADCs.

Since the early 2000s, substantial effort has been directed toward developing technologies to increase linker stability, particularly through alternative types of conjugation chemistry (Supplementary Figure S1, available at <https://doi.org/10.1016/j.annonc.2026.04.008>).¹⁰ In addition, site-specific conjugation strategies have enabled greater linker stability by using positions less prone to deconjugation and more stable chemistry.¹¹ These approaches are often, but not invariably, associated with increased homogeneity and higher systemic ADC exposure. ADCs with more stable linkers have repeatedly shown better therapeutic index preclinically, and many programs have advanced into the clinic based on this promise. Yet, clinical evidence to date does not support enhanced linker stability as a standalone driver of improved therapeutic index in patients.

In this article, we discuss i) types of linker instability, ii) the evaluation of linker stability using *in vitro* assays and *in vivo* pharmacokinetic (PK) analyses and the limitations of pre-clinical models in predicting free payload exposure in humans, iii) our understanding of ADC disposition, and iv) the clinical implications of altering ADC linker stability on clinical maximum tolerated dose (MTD), efficacy, and tolerability.

TYPES OF LINKER INSTABILITY

The clinical safety and efficacy of an ADC are closely tied to the stability of the linker, which influence when, where, and how the cytotoxic payload is released.³ It is important to distinguish linker cleavability from linker instability. Linker cleavability depends on specific environmental conditions intended for payload release (i.e. enzymes, reducing conditions, pH changes) at tumor site, whereas linker instabilities are defined as spontaneous transformations in circulation. In some cases, both processes can overlap and generate the same species. Notably, noncleavable linkers (i.e. linkers that require full ADC catabolism to release biologically active payload-containing species) are not necessarily stable: for example, MCC-DM1 (e.g. used in trastuzumab emtansine) and MC-MMAF (e.g. used in belantamab mafodotin) show marked instabilities,^{12,13} despite being categorized as non-cleavable linkers.

Linker instability (Figure 1)^{3,14} can be classified into linker-drug instability (instability between the linker and the payload) and antibody-linker instability (instability between the antibody and the linker).³ Both instabilities result in reduced DAR over time but through the formation of distinct molecular species. ADCs with linker-drug instability release free payload directly into circulation where it may cause toxicity to normal tissue while also contributing to antitumor activity. ADCs with antibody-linker instability, instead, release the whole drug-linker, which generally requires further metabolism to exert toxic or antitumor effects.

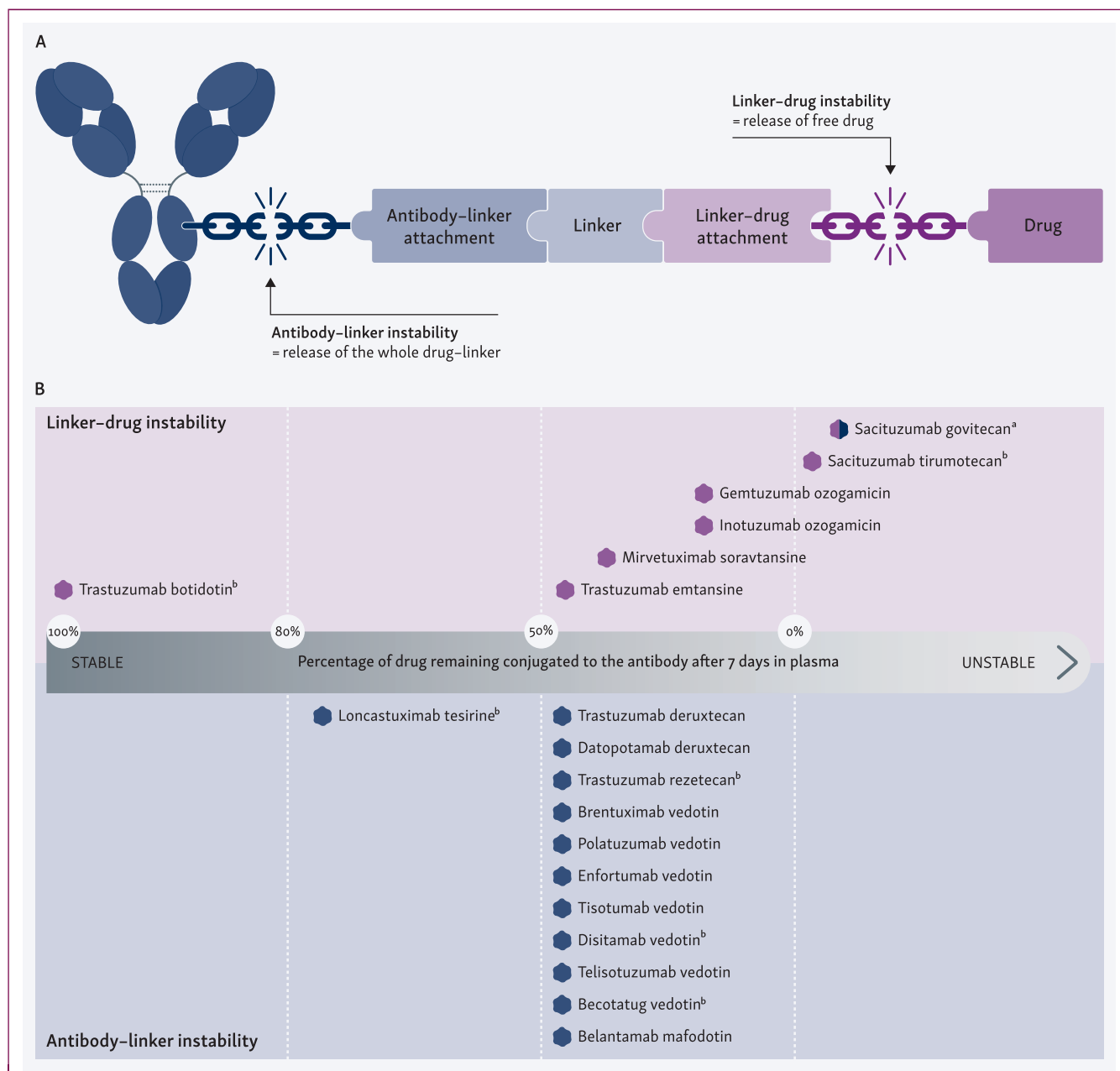


Figure 1. Representation of antibody–drug conjugate (ADC) linker instabilities. (A) ADC instabilities can be divided into antibody-linker instability and linker-drug instability. In recent years, considerable attention has been focused on improving linker stability (Supplementary Figure S1, available at <https://doi.org/10.1016/j.annonc.2026.04.008>). (B) Continuum of stability for approved ADCs, adapted from Colombo et al.³ In blue, ADC with antibody-linker instability, in purple ADCs with linker-drug instability. In addition to linker instabilities, half-life of ADCs is also impacted by intrinsic clearance rates of the different ADC species. Several ADCs feature the same type of instabilities, because of the common chemistry used in the linker. ADCs using vedotin (seven approved ADCs), mafodotin (one approved ADC), deruxtecan, (two approved ADCs), and rezetecan (one approved ADC) drug-linkers feature the same ‘MC’ maleimide, known to deconjugate ~50% after 7 days in plasma.¹⁴ A limited number of ADCs feature prominent linker-drug instability, releasing free payload directly into the bloodstream. These include approved ADCs bearing CL2A and pyrimidine-CL2A linkers (sacituzumab govitecan and sacituzumab tirumotecan, respectively), disulfide (mirvetuximab soravtansine, gemtuzumab ozogamicin, and inotuzumab ozogamicin), and thio-maleimide (trastuzumab emtansine).

NMPA, National Medical Products Administration.

^aSacituzumab govitecan features both antibody-linker and linker-drug instabilities, with the latter more rapid (antibody-linker instability half-life of ~7 days versus linker-drug instability half-life of ~1 day in plasma).

^bCurrently approved by Chinese NMPA but not by the Food and Drug Administration.

Widely employed maleimide linkers are known to undergo deconjugation and subsequently re-conjugate to plasma thiols, primarily to albumin.¹³ Clinical PK and disposition of albumin conjugates have not been reported but warrant investigation given the prevalence of this type of

conjugation. Albumin half-life, and by extension the half-life of albumin conjugates, varies significantly across species (endogenous albumin half-life ~1.5 days in mice, ~2 days in rats, ~6 days in monkeys, ~20 days in humans), making estimation based on preclinical species unreliable.¹⁵

Table 1. Overview of pharmacokinetic analytes and assays for antibody–drug conjugates (ADCs): information provided, limitations, and key considerations

Analyte/assay	What information can it provide?	What information can it not provide?	Considerations
Total antibody (TAb)	- Measurement of all species containing the antibody, regardless of DAR and including unconjugated antibody (DAR = 0) - Target-mediated drug disposition, especially during dose escalation - How ADC design influences overall antibody clearance	- Naked antibody PK - Whether the ADC is intact or not - Linker stability - DAR over time - Antibody-conjugated drug exposure (acDrug) - Payload delivery	- For ADCs, it is common that TAb half-life differs from that of the PK of the parent unconjugated antibody, as conjugation may impact nonspecific clearance. - TAbs include fully deconjugated (DAR0) antibody, which only appears after complete payload loss. <i>Common mistake:</i> - Similar TAb and ADC PK concentrations are often misinterpreted as an indicator of linker stability
Antibody-conjugated drug (acDrug)	- Measurement of drug that is still linked to the antibody - Rate of payload loss from antibody (when compared with TAb)	- Distinguish antibody-linker vs. linker-drug instability - Determine DAR distribution - Detect albumin–drug conjugates	- acDrug/TAb molar ratio approximates average DAR at any given time point. - acDrug decays faster than TAb for unstable ADCs. - If anti-payload antibody is used for immune-precipitation, it may also capture payload conjugated to albumin or other thiol-containing species
ADC (also called total ADC)	- Overall ADC half-life and exposure - It measures payload conjugated antibody species (i.e. net of all the antibody species with at least one drug attached)	- Assess linker stability directly - Distinguish antibody-linker vs. linker-drug instability - Detect DAR0 species (naked antibody)	- Assay includes any payload conjugated antibody species (DAR$\geq$1). - Differences between TAb and ADC levels become apparent only after formation of DAR0 species - Less informative than acDrug <i>Common mistake:</i> - Similar ADC and TAb PK concentrations are often misinterpreted as an indicator of linker stability
Free payload	Systemic exposure and apparent half-life of payload released from the antibody	- Detect antibody-linker instability - Distinguish release via ADC catabolism vs. linker instability - Intrinsic half-life of the payload	- Free payload level reflects both rate of formation (instability + catabolism) and rate of clearance, therefore it is not a measure of linker stability. - Preclinical species consistently underestimate free payload vs. humans. <i>Do not</i> use animal data to directly predict clinical payload exposure. <i>Common mistake:</i> - Low levels of free payload in plasma, especially in preclinical species, is often misinterpreted as an indicator of linker stability
DAR over time	- How ADC DAR average changes over time <i>In vitro</i>: it directly quantifies linker instability (both types) and changing DAR average <i>In vivo</i>: it reflects both linker instability and preferential clearance of higher DAR species	<i>In vivo</i>: distinguish deconjugation from DAR-dependent clearance	- Rarely reported in clinical studies <i>In vitro</i>: it may overestimate stability, as active plasma enzymes or reactive species may be absent <i>ex vivo</i>, limiting <i>in vivo</i> relevance
<i>In vitro</i> ADC stability assay	- Measurement of free payload released <i>in vitro</i>, typically expressed as a percentage of the total releasable payload from the ADC. - Commonly carried out <i>ex vivo</i> using serum or plasma from different species (e.g. human, monkey, rat, mouse)	Detect any instability or biotransformation that does not release the intact free payload (or specific analytes measured in the assays)	- Useful to assess linker-drug stability - It requires knowledge of all the released payload species to fully assess linker stability - In plasma or serum matrices, it may overestimate stability, as active enzymes or reactive species may be reduced <i>ex vivo</i>, limiting <i>in vivo</i> relevance. <i>Common mistake:</i> - Low levels of free payload is often misinterpreted as an indicator of linker stability
Albumin–drug conjugates	Apparent half-life and exposure of albumin–drug conjugate formed through antibody deconjugation and subsequent re-conjugation of drug-linker to albumin	Direct measurement of overall linker stability	- Only relevant for ADCs with drug-linkers that can undergo deconjugation from antibody and re-conjugation to albumin - Not yet reported in clinical trials

DAR, drug-to-antibody ratio; PK, pharmacokinetic.

The majority of currently approved ADCs feature antibody-linker instability. Notably, 11 of 19 approved ADCs incorporate MC-based linkers, which detach ~50% of their drug-linker moiety over a week in plasma (Figure 1B).^{3,14} Sacituzumab govitecan (SG) features both antibody-linker instability and linker-drug instability. Ozogamicin ADCs have two points of instability in their linker (i.e. hydrazone and disulfide)³ resulting in ADCs with relatively low stability.¹⁶ Thus far, loncastuximab tesirine is the maleimide-based approved ADC with the most stable linker (MP-PEG₈-VA-PABC-SG3199), losing ~20% of the conjugated drug over a week in plasma, similar to other ADCs with MP-PEG₈ linkers.¹⁷ Trastuzumab botidotin represents the first approved stable ADC (both linker-drug and antibody-linker) and is associated with a high incidence (>90%) of ocular toxicity, as discussed below.¹⁸

PRECLINICAL AND CLINICAL EVALUATION OF LINKER INSTABILITY AND THE LIMITATIONS OF PRECLINICAL MODELS IN PREDICTING HUMAN PAYLOAD EXPOSURE

Linker stability can be evaluated using *in vitro* assays and through preclinical and clinical *in vivo* PK measurements, each with specific advantages and limitations (Table 1). Overreliance on these methods without a nuanced understanding of what they measure has often led to misinterpretation of linker stability and confusion interpreting clinical data.

Sequential detachment of a payload or a drug-linker converts an ADC from higher to lower DAR, potentially leading to antibody species with no payload attached. These chemical transformations are well understood preclinically, yet they remain underappreciated and are not routinely characterized in the clinic. For maleimide-based ADCs, the rate of deconjugation is impacted by the maleimide structure and the conjugation site (Supplementary Figure S1, available at <https://doi.org/10.1016/j.annonc.2026.04.008>). Payload and linker components more distant from the maleimide do not significantly impact deconjugation.^{14,19} *In vitro*, only the intrinsic deconjugation rate is apparent (closed system). In contrast, *in vivo* (open system) the decrease in average DAR reflects both the intrinsic deconjugation rate and the more rapid clearance of high-DAR species, if there is a marked DAR-dependent clearance due to increased hydrophobicity associated with higher DAR.^{20,21} For example, similar deconjugation rates were observed for MMAE (more hydrophobic) and MMAF (less hydrophobic) ADCs *in vitro*, but clinical samples showed faster DAR loss for MMAE ADCs, reflecting both deconjugation and preferential clearance of high-DAR species.²¹

In preclinical and clinical settings, the most robust method to evaluate DAR over time is by directly measuring ADC DAR in serum or plasma using mass spectrometry (MS). (Figure 2A,²¹⁻²⁶ Table 1). In preclinical models, this is readily achieved through generic immunocapture and subsequent MS analysis.^{14,27} In clinical settings, the analysis is complicated by the presence of native immunoglobulins, requiring selective ADC capture followed by MS analysis.²¹

As a result, direct measurement of linker instability in patients remains limited thus far.

An indirect approach to understanding DAR changes is to compare total antibody (TA_b) and antibody-conjugated drug (acDrug) PK measurements (Figure 2B, Table 1). TA_b assays quantify all antibody species, including unconjugated antibodies, regardless of the extent of conjugation. In contrast, acDrug assays measure the amount of payload bound to the antibody.²⁸ The molar ratio of acDrug to TA_b at any time point approximates the DAR. At the intersection of TA_b and acDrug curves, DAR equals 1. Clinical PK of vedotin-based ADCs provide illustrative examples.²³ However, interpatient differences and inherent assay variability can limit the precision of assessing the ratio between these analytes.

Comparisons of ADC and TA_b PK are sometimes used to claim that an ADC is stable, but differences between TA_b and ADC levels become apparent only after formation of DAR0 species (Figure 2C, Table 1).²⁸ Consequently, differences are more likely to appear for ADCs with lower initial DAR (e.g. DAR2 or DAR4). For high-DAR ADCs (e.g. DAR8), similar ADC and TA_b PK concentrations could give a false impression of stability, because of the limited transformation from DAR8 to DAR0 over the course of the assay (Figure 2C, Table 1). For example, initial studies of trastuzumab deruxtecan (T-DXd; T-MC-GGFG-DXd, DAR8) erroneously reported that 'PK profiles of T-DXd and TA_b were similar, indicating that the linker is stable in circulation.'²⁹ Only more recently, in-depth analyses highlighted that T-DXd loses 50% of the conjugated drug over 1 week and 75% over 3 weeks (Figure 2A) through maleimide deconjugation.^{24,30} T-DXd instability,^{24,30} fully anticipated based on its linker structure, is inherent to all ADCs featuring MC-based or MCC-based linkers and native interchain cysteine conjugations. Beyond T-DXd, other examples illustrate the risk of comparing ADC and TA_b to evaluate stability.³¹⁻³⁴

Reliance on *in vitro* measurements of free payload released from ADCs in plasma to understand linker stability is another common mistake (Table 1). This assay only captures linker-drug instability and does not directly detect antibody-linker instability or transformations that do not involve the release of the intact free payload.⁹ For example, it has been erroneously suggested that T-DXd is stable in plasma because of the low DXd percentage released *in vitro* over 21 days.³⁵ Several other studies have reached similar inaccurate conclusions about ADC stability based on free payload released *in vitro*,^{33,36-40} overlooking other important degradation pathways (e.g. antibody-linker stability).

Using plasma payload levels relative to ADC (or TA_b) levels from clinical samples as a surrogate for linker stability can also be misleading (Table 1), as the amount of free payload in plasma reflects both its rate of generation (via linker instability and ADC catabolism) and its clearance (via small-molecule elimination routes). For example, for SHR-A1811, SHR-A1904, and HS-20093, low plasma payload concentrations have been incorrectly interpreted as evidence of high linker stability.^{34,41,42} These ADCs are not stable: they employ an MC-based linker that is susceptible to deconjugation, similar to T-DXd and many other ADCs (for structural comparison of T-DXd, SHR-A1811, SHR-A1904, and

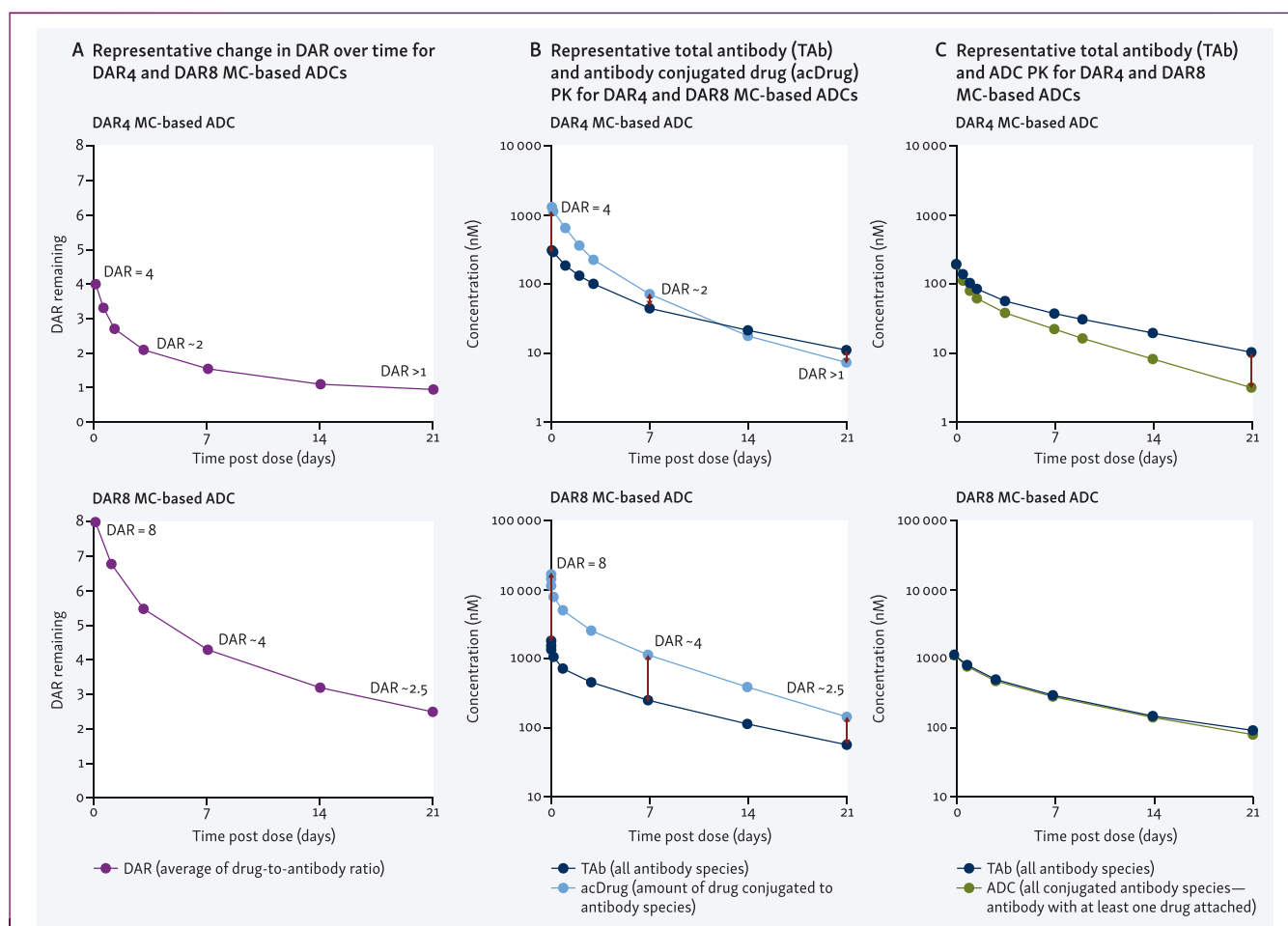


Figure 2. Representative measurements of *in vivo* drug-to-antibody ratio (DAR) over time, total antibody, antibody-conjugated drug, and total antibody–drug conjugate (ADC) for common DAR4 and DAR8 MC-based ADCs illustrate that different assays can provide different impressions of stability. (A) Change in average DAR over time for DAR4 and DAR8 MC-based ADCs. (B) Antibody-conjugated drug (acDrug) for DAR4 and DAR8 MC-based ADCs. acDrug decays faster than total antibody (Tab), due to deconjugation. For unstable ADCs and ADCs with marked preferential clearance of high-DAR species, the concentration of acDrug decreases more rapidly than Tab.²² For DAR4, it is more likely that the two lines cross, highlighting the ADC average DAR of 1. This is less likely for MC-based DAR8 ADCs, owing to the greater number of deconjugation events required. (C) Total antibody (Tab) versus ADC PK for DAR4 and DAR8 MC-based ADCs. Because no significant amount of naked antibody (DAR0) is present for DAR8 ADCs, the two curves tend to overlap more than for DAR4 ADCs. Representative data extracted from the references.^{21–26} MC, maleimidocaproyl; PK, pharmacokinetic.

HS-20093, see [Supplementary Figure S2](https://doi.org/10.1016/j.annonc.2026.04.008), available at <https://doi.org/10.1016/j.annonc.2026.04.008>).

Finally, free payload measurement in preclinical species is even more misleading. Across different payload classes and ADC designs (stable or unstable), free payload exposure is significantly underestimated in preclinical models (Figure 3,^{24–26,33,39,43–55} [Supplementary Figures S3–S5](https://doi.org/10.1016/j.annonc.2026.04.008), available at <https://doi.org/10.1016/j.annonc.2026.04.008>). This includes nonhuman primates (NHPs), which are often considered the only pharmacologically relevant species for ADCs. Although the exact reasons are unclear, a possible explanation is that clearance of released free payload (a small molecule) often scales with body size across species, with faster metabolism and excretion processes in smaller animals.⁵⁶ In contrast, ADC clearance rates are more comparable across species.⁵⁷ As a result, unlike in mice, rats, and nonhuman primates, payload concentrations relative to ADC (or Tab) are higher in humans, maintaining pharmacologically relevant levels for an extended period of time.^{3,58} These

interspecies differences are often underappreciated and can lead to flawed interpretations of ADC stability and safety. Numerous reports have incorrectly argued that low free payload concentrations in preclinical species are evidence of ADC stability and a predictor of improved clinical safety.^{31,37,40,59–65} The underestimation of free payload exposure in preclinical species and translational limitations across species (Figure 3) should be borne in mind when optimizing ADCs. Analyses of DAR, Tab, acDrug, and ADC PK in tumor-bearing mouse models (for efficacy studies)⁵⁷ or healthy animals (commonly rodent and NHP for tolerability studies)⁴³ provide valuable information on how ADC design influences stability and clearance, and ultimately efficacy and tolerability. However, it is imperative to realize that preclinical models substantially underestimate free payload exposure relative to human data (Figure 3). This discrepancy further complicates the translation of ADCs, which are already challenged by their inherent complexity and multi-component nature.

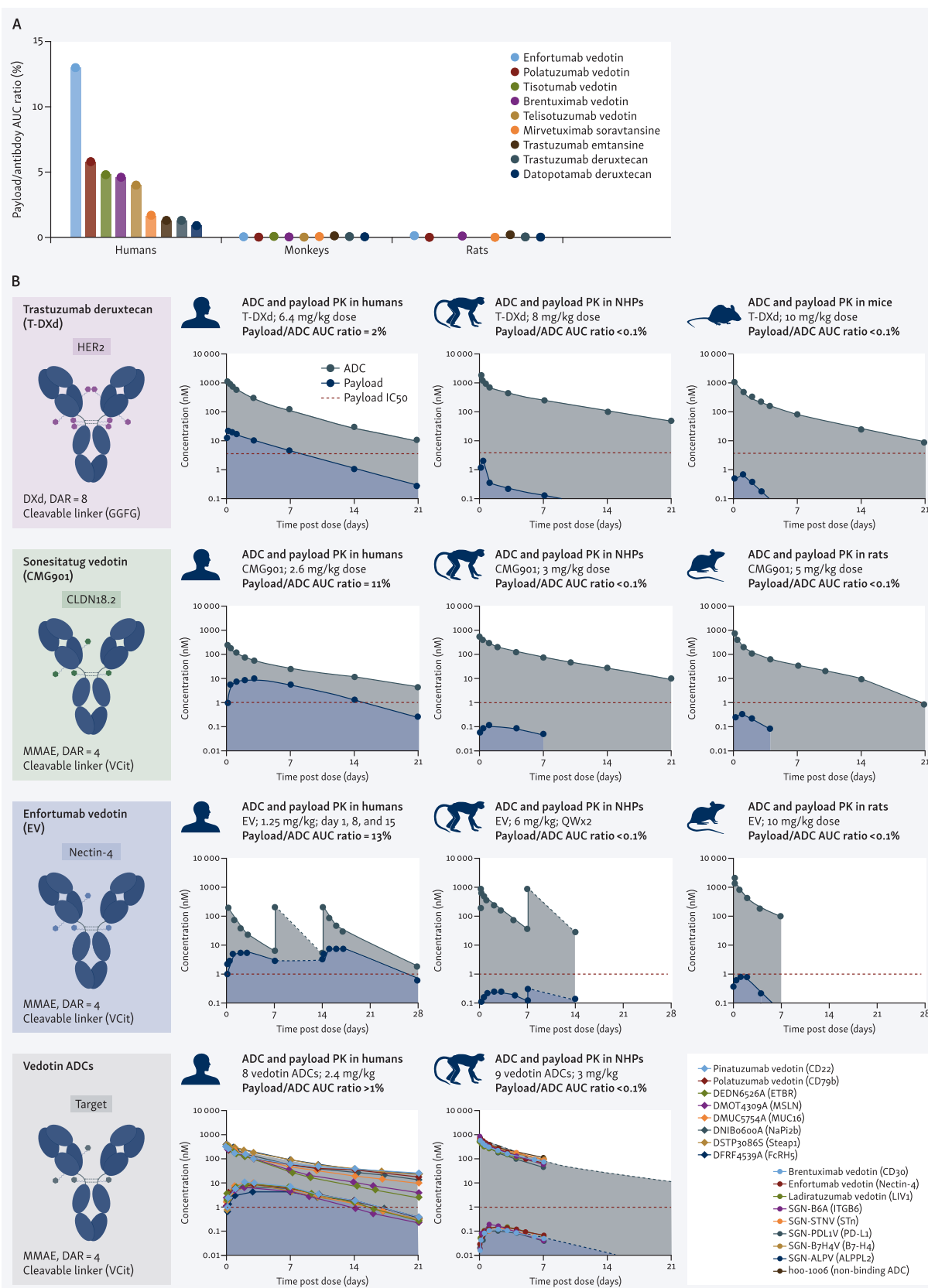


Figure 3. Comparisons of free payload to ADC exposure in different species. (A) ADC payload exposure in humans is significantly higher than it is in preclinical species, including NHPs, highlighting how the contribution of circulating payload to both efficacy and tolerability is likely underestimated in all the preclinical models. (B) Representative PK curves for a deruxtecan ADC, a vedotin ADC, and a vedotin ADC dosed QW, across humans, NHPs, and rodents. (C) PK of multiple vedotin ADCs dosed at 2.4 mg/kg in humans and 3 mg/kg in NHPs highlights consistent underestimation of payload exposure in preclinical animal models. Data were extracted from the references^{24-26,33,39,43-55}. For additional examples, see [Supplementary Figures S3-S5](https://doi.org/10.1016/j.annonc.2026.04.008), available at <https://doi.org/10.1016/j.annonc.2026.04.008>. ADC, antibody–drug conjugate; AUC, area under the curve; NHP, nonhuman primate; PK, pharmacokinetic; QW, once a week.

UNDERSTANDING ADC DISPOSITION

ADCs, like all antibody-based therapeutics, are cleared almost exclusively through intracellular lysosomal degradation after cellular uptake via non—receptor-mediated and receptor-mediated pathways.^{66,67} For antibodies and ADCs, only a small fraction of the injected dose (often cited as less than 1%) typically reaches the tumor site in patients.³ Antibody conjugation can result in higher clearance than for the parent naked antibody. For example, an analysis of the clinical PK of 14 different trastuzumab-based ADCs compared with unconjugated trastuzumab, highlights a strong dependence on drug-linker properties (Supplementary Figure S6 and Supplementary Table S1, available at <https://doi.org/10.1016/j.annonc.2026.04.008>). Relative to its unconjugated antibody, faster ADC clearance is driven by increased nonspecific uptake rather than target-mediated disposition, as both share the same antibody, implying that an even smaller fraction of the injected ADC dose is expected to reach the tumor. Clinical biodistribution data for radiolabeled ADCs in patients are extremely limited,⁶⁸⁻⁷¹ as majority of studies used radiolabeled antibodies⁷² rather than radiolabeled ADCs. This gap needs to be addressed to improve our understanding of ADC disposition. Among the few available studies, the biodistribution of CMD-193 ADC (anti-Lewis Y antibody hu3S193 conjugated to calicheamicin)⁷¹ and its parent antibody hu3S193⁷³ is particularly illustrative. In patients, radiolabeled CMD-193 (ADC) exhibited markedly higher systemic clearance, increased hepatic uptake, and minimal tumor localization compared with its radiolabeled parent antibody.^{71,73} Importantly, one patient participated in both studies, allowing a direct and clear comparison.^{71,73}

Although uptake and lysosomal degradation of an antibody yields small peptides and amino acids with no detrimental impact on the cell viability,⁷⁴ off-tumor catabolism of an ADC ultimately releases a cytotoxic payload. With certain exceptions, ADC clinical toxicities and MTDs are predominantly payload related, further highlighting the importance of ADC nonspecific normal tissue disposition and the appropriate payload selection.⁷⁵⁻⁷⁸ This is well exemplified by trastuzumab-based ADCs (including trastuzumab conjugated to emtansine, duocarmazine, deruxtecan, rezetecan, brentidecan, pamirtecane, bultecan, vedotin, envedotin, botidotin, imbotolimod, and others), which exhibit distinct drug-linker—dependent clinical toxicities and MTDs despite the common HER2-targeting antibody (Supplementary Table S1, available at <https://doi.org/10.1016/j.annonc.2026.04.008>).

IMPACT OF LINKER STABILITY ON ADC AND PAYLOAD DISPOSITION

Given that the vast majority of an ADC-injected dose distributes to normal tissues, increased linker stability may result in higher and prolonged exposure of normal tissue to the intact ADC (Supplementary Figure S7, available at <https://doi.org/10.1016/j.annonc.2026.04.008>), which may cause unexpected toxicities. Circulating payload originates not only from linker instability but also from ADC catabolism, therefore levels of free payload in circulation may be similar between more-stable

and less-stable ADCs.²¹ Indeed, approximately four-fold higher antibody-conjugated drug exposure (acDrug area under the curve) was observed in a phase 1 trial for the more-stable iladatumab vedotin compared with the less-stable polatumab vedotin; however, free payload exposure was similar for these two ADCs (Supplementary Figure S8, available at <https://doi.org/10.1016/j.annonc.2026.04.008>).^{79,80} A similar trend was observed for the MUC16-targeting less-stable sofituzumab vedotin compared with the more stable DMUC4064A^{81,82} and the cMET-targeting less-stable telisotuzumab vedotin compared with the more stable MYTX-011,⁸³ although in this last case, the antibodies feature different paratopes.

ADCs with more-stable linkers do not necessarily have better PK (in terms of TAb, ADC, or acDrug). For example, Nectin-4 9MW2821 (employing a stable Cys-bridged maleimide-PEG-VC-PABC-MMAE) showed higher clearance than enfortumab vedotin (employing MC-VC-PABC-MMAE linker, stochastic conjugation), a less-stable Nectin-4 ADC (Supplementary Figure S9, available at <https://doi.org/10.1016/j.annonc.2026.04.008>).^{53,84} However, the antibody sequences of 9MW2821 and EV are distinct, which may also contribute to the observed differences. Rapid ADC clearance may result from factors beyond linker stability, including normal target expression, antibody format, or increased hydrophobicity associated with the antibody sequence, drug-linker type, DAR, and conjugation. For example, despite having a stable linker, bispecific ADC izalontamab brengitecan has a half-life of 1-1.5 days,⁸⁵ possibly due to its antibody format.

The ADC literature routinely implies that most free drug in circulation originates from linker instability.¹ This overlooks the reality of ADC disposition, which predominantly involves cellular uptake followed by catabolism and payload release, rather than premature payload release in circulation. Linker-payload properties, such as hydrophobicity or charge, may impact nonspecific uptake and clearance.^{21,86} For unstable ADCs, free payload can also be generated directly in circulation via linker-drug instability or, in case of antibody-linker instability, the whole drug-linker can be transferred to albumin and then be metabolized as albumin conjugate.³

THE IMPLICATIONS OF ADC LINKER STABILITY ON CLINICAL MTD, EFFICACY, AND TOLERABILITY

Increased linker stability does not necessarily increase MTD

Many toxicities are specific to the payload, rather than the target, and these are commonly referred to as platform toxicities.³ Increased ADC stability does not necessarily translate into improved clinical MTDs, expressed as normalized cytotoxin content ($\text{Dose}_{\text{ADC}} \times \text{DAR} \times \text{MW}_{\text{payload}} / \text{MW}_{\text{ADC}}$), a useful metric to compare ADCs with different DAR.^{3,75} Exceptions include ADCs which may exhibit lower MTDs due to marked on-target off-tumor toxicities. For instance, several TROP2 TOPO1i ADCs reach ~50% of the clinically tolerated dose of other ADCs that employ the same drug-linker technology (Supplementary Figure S10,

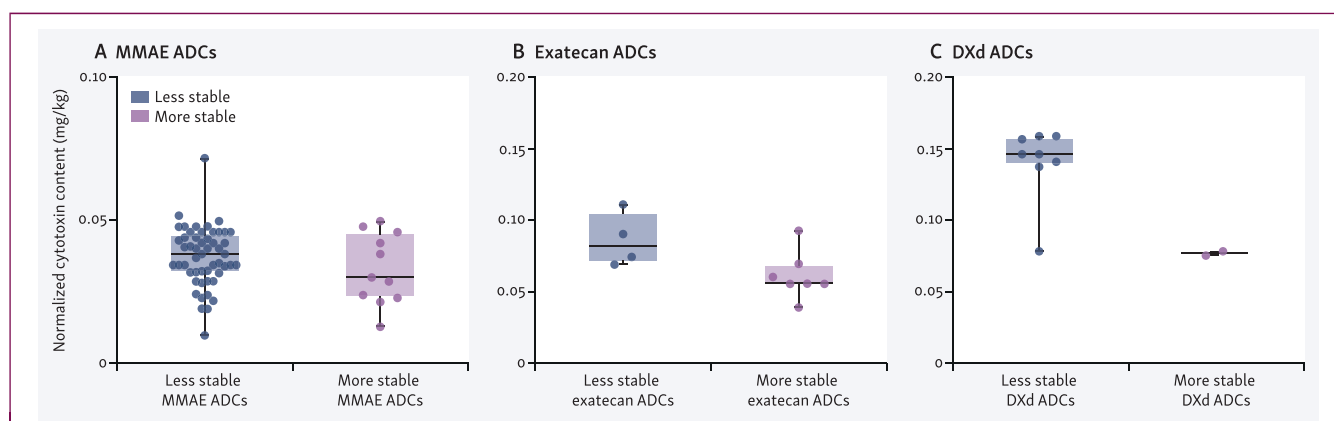


Figure 4. Clinical dose (established RP2D, MTD, MAD, and dose expansions doses, see [Supplementary Table S2](https://doi.org/10.1016/j.annonc.2026.04.008), available at <https://doi.org/10.1016/j.annonc.2026.04.008> for details and references) of more-stable and less-stable MMAE, exatecan, and DXd ADCs expressed as normalized cytotoxin content ($\text{Dose}_{\text{ADC}} \times \text{DAR} \times \text{MW}_{\text{payload}} / \text{MW}_{\text{ADC}}$). (A) Comparison of clinical dose for MMAE ADCs with standard MC-based vedotin linker (blue) and more stable MMAE ADCs (pink); (B) comparison of clinical doses for MC-based exatecan ADCs (blue) and more stable exatecan ADCs (pink); (C) comparison of clinical doses for MC-based DXd ADCs (blue) and more-stable DXd ADCs (pink).

ADC, antibody–drug conjugate; DXd, deruxtecan; MAD, maximum administered dose; MTD, maximum tolerated dose; RP2D, recommended phase 2 dose.

available at <https://doi.org/10.1016/j.annonc.2026.04.008>). These lower clinically achievable doses are likely driven by TROP2-mediated toxicities, such as stomatitis. Notably, SG, which incorporates a relatively unstable linker (half-life <24 h),⁸⁷ is not associated with stomatitis.⁸⁸ Rather, the SG toxicity profile largely mirrors that of its payload, with prominent neutropenia and gastrointestinal toxicity.⁸⁹

An analysis of MMAE ADCs with reported clinical data shows that both more-stable and less-stable ADCs tend to achieve similar doses (Figure 4A),³ expressed as normalized cytotoxin content.^{3,75} For other payload classes, fewer examples are available for direct comparison. For exatecan ADCs, clinical doses reported to date do not appear to differ markedly between more-stable and less-stable constructs (Figure 4B), consistent with the trend observed for MMAE; however, data are not as mature. Among DXd ADCs, several conjugates exhibiting antibody-linker instability (including MC-based T-DXd, HER3-DXd, I-DXd, R-DXd, DS-6157, and FDA022) reach doses (normalized cytotoxin content) approximately two-fold higher than those achieved by two more-stable DXd ADCs, GQ1005, and JSKN003 (Figure 4C). Details for stable and less-stable MMAE, exatecan, and DXd ADCs included in the analysis are provided in [Supplementary Table S2](https://doi.org/10.1016/j.annonc.2026.04.008) (available at <https://doi.org/10.1016/j.annonc.2026.04.008>). Given the large number of ADCs with same payloads that have entered clinical development, future clinical data may further elucidate the relationship between linker stability and tolerability.

Increased ADC linker stability may result in unexpected toxicities

Emerging clinical data for ADCs that share the same payload and target but different linker stability provide important insights into the impact of linker design. For example, traditional stochastic MMAE ADCs, characterized by

antibody-linker instability ($\geq 50\%$ DAR loss in plasma over 1 week), are associated with neutropenia and peripheral neuropathy, toxicities often attributed to ‘premature payload release’ (although this explanation clearly oversimplifies ADC disposition, as discussed). Consequently, more stable MMAE ADCs have been proposed to improve upon earlier vedotin constructs.^{90,91} Despite intense efforts over the past decade to develop MMAE ADCs with increased linker stability, none has been approved to date (Figure 1). Notably, several MMAE-based ADCs with increased linker stability have been associated with higher incidence of ocular toxicities in the clinic, including dry eyes, blurred vision, and corneal changes (Table 2).^{44,45,48,55,81,82,84,92–102}

Ocular toxicity is a complex side-effect that can significantly impact patient quality of life (QoL). Although pathophysiology of ADC-related ocular events has not been fully elucidated, ocular toxicities are attributed to both on-target and off-target mechanisms and are often payload related.^{103,104} In particular, ADCs bearing either MMAF or DM4 payloads have consistently been associated with ocular toxicities, irrespective of target, clearly pointing to payload-dependent nonspecific mechanisms.¹⁰⁵ Clinical experience with pyrrolobenzodiazepine-based ADCs has included reports of photophobia, among other toxicities.^{106,107} TOPO1i ADCs have generally showed a low incidence of ocular toxicity,⁷⁸ with rare exceptions (e.g. datopotamab deruxtecan, likely related to TROP2-expression in the cornea).⁵² In contrast, MMAE ADCs show more variable ocular toxicity profiles, influenced by both linker stability and target expression, and therefore warrant more in-depth consideration. MMAE ADCs with moderate linker stability (e.g. classic stochastic vedotin) rarely cause significant ocular toxicity, except when the target is expressed in the eye, as seen with tisotumab vedotin, where tissue factor is present in the conjunctiva.^{3,105} Unexpectedly, MMAE ADCs with increased linker stability showed moderate to very high incidence of ocular side-effects. The incidence of ocular toxicities among

Table 2. A summary of vedotin (MC-VCit-PABC-MMAE) and vedotin-like (VCit-PABC-MMAE) ADCs with available clinical data across common targets, along with reported ocular toxicities

VCit-PABC-MMAE ADCs with lower antibody-linker stability							
ADC	Target	Antibody	Linker-drug	Conjugation	DAR	Ocular toxicity	Reference
Trastuzumab vedotin (MRG002)	HER2	Trastuzumab	MC-VCit-PABC-MMAE	Stochastic Cys conjugation	4	No ocular adverse effects reported	92
Disitamab vedotin	HER2	Disitamab	MC-VCit-PABC-MMAE	Stochastic Cys conjugation	4	Ocular adverse events (grade 1-2): 17/108 (15.4%) in real-world setting ¹	93
ALT-P7	HER2	HM2 [trastuzumab with C-terminal metal ion-binding motifs (ACGHA)]	MC-VCit-PABC-MMAE	Site-specific Cys conjugation ²	2	No ocular AEs reported	94
Sofituzumab vedotin	MUC16	Sofituzumab	MC-VCit-PABC-MMAE	Stochastic Cys conjugation	4	No ocular AEs reported	81
Polatuzumab vedotin	CD79b	Polatuzumab	MC-VCit-PABC-MMAE	Stochastic Cys conjugation	4	1.2% any grade blurred vision	45
CMG901/AZD0901	CLDN18.2	Sonesitatumab	MC-VCit-PABC-MMAE	Stochastic Cys conjugation	4	No ocular AEs reported	55
ATG-022	CLDN18.2	N/A	MC-VCit-PABC-MMAE	Stochastic Cys conjugation	4	No ocular AEs reported	95
LM-302	CLDN18.2	Tecotabart	MC-VCit-PABC-MMAE	Stochastic Cys conjugation	4	No ocular AEs reported	96
RC118	CLDN18.2	Ciletatumab	MC-VCit-PABC-MMAE	Stochastic Cys conjugation	4	No ocular AEs reported	97
Enfortumab vedotin	Nectin-4	Enfortumab	MC-VCit-PABC-MMAE	Stochastic Cys conjugation	4	PADCEV package insert contains an ocular disorder warning 24% any grade dry eye, 0.7% grade ≥ 3 in EV-301 as monotherapy	44
Telisotuzumab vedotin	cMET	Telisotuzumab	MC-VCit-PABC-MMAE	Cys conjugation ³	3.1	EMRELIS package insert contains an ocular surface disorder warning 15% any grade blurred vision, 1.2% grade ≥ 3 11% any grade keratitis, 0.6% grade ≥ 3 5% any grade dry eye	48
Tisotumab vedotin	Tissue factor	Tisotumab	MC-VCit-PABC-MMAE	Stochastic Cys conjugation	4	TIVDAK package insert contains a boxed warning for potential severe ocular toxicities 32% any grade conjunctival adverse reactions, 0 grade ≥ 3 21% any grade corneal adverse reactions, 3.2% grade ≥ 3 21% any grade dry eye, 0 grade ≥ 3	46
VCit-PABC-MMAE ADCs with higher antibody-linker stability							
ADC	Target	Antibody	Linker-drug	Conjugation	DAR	Ocular toxicity	Reference
DP303c	HER2	Trastuzumab	PEG3-VCit-PABC-MMAE	Site-specific conjugation at Gln 295 (Q295)	2	87% any grade corneal disease, 2% grade ≥ 3 62% any grade blurred vision, 16% grade ≥ 3 57% any grade dry eye, 6% grade ≥ 3 19% any grade eye pain, 1% grade ≥ 3 18% any grade increased lacrimation, 0 grade ≥ 3 18% any grade photophobia, 0 grade ≥ 3	98
DMUC4064A	MUC16	Same CDR as sofituzumab, thiomab version	MC-VCit-PABC-MMAE	Site-specific Cys conjugation	2	35% any grade blurred vision, 3% grade ≥ 3 17% any grade dry eye, 0 grade ≥ 3 14% any grade keratitis, 6% grade ≥ 3	82

Continued

Table 2. (Continued)

VCit-PABC-MMAE ADCs with higher antibody-linker stability							
ADC	Target	Antibody	Linker-drug	Conjugation	DAR	Ocular toxicity	Reference
Iladatuzumab vedotin	CD79b	Iladatuzumab, same CDR as polatuzumab, Thiomab version	MC-VCit-PABC-MMAE	Site-specific Cys conjugation	2	Phase 1 (dose escalation and dose expansion as monotherapy and in combination with rituximab): 66 ocular AEs occurred in 28/60 patients (47%). Grade 1-2 (39%) and grade 3 (8%). 35% any grade blurred vision, grade ≥ 3 not reported 28% any grade corneal deposits, grade ≥ 3 not reported 10% any grade keratitis, grade ≥ 3 not reported	⁸⁰
CRB-701 (SYS6002)	Nectin-4	SWY2001-Ab1	PEG3-VCit-PABC-MMAE	Site-specific conjugation at Gln 295 (Q295)	2	Western study: 34% any grade dry eye or keratitis, 5% grade ≥ 3 Chinese study: 57% any grade corneal disorder, 6% grade ≥ 3 38% any grade dry eye, 0 grade ≥ 3	⁹⁹
9MW2821 ⁴	Nectin-4	Bulumtatug	IDM-VCit-PABC-MMAE	Bridged Cys conjugation	4	Not reported	⁸⁴
EO-3021 (SYSA1801)	CLDN18.2	SYJS001-mAb	PEG3-VCit-PABC-MMAE	Site-specific conjugation at Gln 295 (Q295)	2	Western study: 19% any grade, 3% grade ≥ 3 Chinese study: 21% any grade dry eye, 0 grade ≥ 3	¹⁰⁰
MYTX-011	cMET	MYTX-011 IgG1 mAb	MC-VCit-PABC-MMAE	Site-specific Cys conjugation	2	At doses ≥ 4 mg/kg 47% any grade blurred vision, 20% grade ≥ 3 41% any grade keratopathy, 18% grade ≥ 3 12% any grade dry eye, 2% grade ≥ 3	¹⁰¹
MRG004a	Tissue factor	TF-mAb	BCN-hydraspase-(VCit-PABC-MMAE) ₂	Glycan engineering (GlycoConnect™)	<4	35% any grade conjunctivitis, 6% grade ≥ 3 22% any grade keratitis, 3% grade ≥ 3 11% any grade blurred vision, 0 grade ≥ 3 10% any grade dry eye, 0 grade ≥ 3	¹⁰²

Greater ocular toxicity appears to be associated with higher antibody-linker stability. This association should be interpreted as correlative rather than causal, as cross-trial comparisons may be confounded by multiple factors beyond linker stability. Nevertheless, clinical and preclinical data suggest that auristatin ADCs with more stable linkers and higher conjugated drug exposure are associated with increased ocular toxicity.

ADC, antibody–drug conjugate; AE, adverse event; BCN, bicyclo[6.1.0]non-4-yne; CDR, complementarity-determining region; DAR, drug-to-antibody ratio; IDM, interchain disulfide maleimide (a hydrolyzable di-substituted maleimide); MC, maleimidocaproyl; PABC, *p*-aminobenzyl carbamate; PEG, polyethylene glycol.

¹Ocular adverse events were not reported in clinical trials nor in the NMPA version of the package insert.

²Presumed to be less stable based on exposed C-terminal cysteine conjugation.

³Hinge stabilized IgG1 with HIC purification of stochastic interchain cysteine conjugation to enrich lower DAR species.

⁴Stable DAR4. Same dose and regimen as enfortumab vedotin, but higher ADC clearance than enfortumab vedotin (see [Supplementary Figure S9](https://doi.org/10.1016/j.annonc.2026.04.008), available at <https://doi.org/10.1016/j.annonc.2026.04.008>).

target-matched pairs of less-stable and more-stable MMAE ADCs is contrasted in Table 2. Clearly, the toxicity of ADCs delivering the same payload to the same target (sometimes even the same antibody) is influenced by linker stability, although not always in the way it has conventionally been interpreted.

In addition to the MMAE ADCs (Table 2), high levels of ocular toxicity have been observed with other auristatin ADCs featuring high linker stability (e.g. trastuzumab botidotin, ARX788, ZV0203).¹⁰⁸⁻¹¹⁰ This general association between ocular toxicity and auristatin ADC stability is likely due to higher acDrug exposure (and consequently higher payload levels) in the eye compared with less-stable ADCs. This interpretation is supported by the analyses of clinical PK (Supplementary Figure S8, available at <https://doi.org/10.1016/j.annonc.2026.04.008>) and by preclinical findings.^{111,112} Ocular toxicity for more stable auristatin ADCs with targets not expressed in the eye (e.g. CD79b) suggests nonspecific mechanisms.¹⁰³ However, when the target is expressed in ocular tissue, increased on-target uptake cannot be ruled out.¹⁰³ Notably, if a stable linker leads to reduced ADC exposure, as in the case of 9MW2821, it may not result in higher ocular toxicity (Supplementary Figure S9, available at <https://doi.org/10.1016/j.annonc.2026.04.008>).⁸⁴

It is also instructive to compare toxicities of DM1 ADCs featuring either a noncleavable MCC linker (MCC-DM1) or a cleavable SPP linker (SPP-DM1). ADCs employing MCC-DM1 (e.g. T-DM1) lose ~50% of the DM1 payload over a week.¹¹³ SPP-DM1 is even less stable than MCC-DM1 (for chemical structures and stability data of MCC-DM1 and SPP-DM1, see Supplementary Figure S11, available at <https://doi.org/10.1016/j.annonc.2026.04.008>).⁶ MCC-DM1 releases DM1 because of linker-drug instability and forms Lys-MCC-DM1 as a major metabolite (nonpermeable but cytotoxic) from cellular catabolism. The SPP-DM1 linker releases DM1 via both linker-drug instability and cellular catabolism. Significant thrombocytopenia was reported for MCC-DM1 ADCs (including trastuzumab emtansine, Debio1562, PCA062, AMG 595, AMG 224, AMG 172), whereas thrombocytopenia was not reported for SPP-DM1 ADCs (MLN2704, lorvotuzumab mertansine, cantuzumab mertansine, bivatuzumab mertansine) (see Supplementary Table S3, available at <https://doi.org/10.1016/j.annonc.2026.04.008>). Two other maytansinoid ADCs with a more stable linker than SPP-DM1 (BAT8001 and TRPH-222) were also associated with severe thrombocytopenia (Supplementary Table S3, available at <https://doi.org/10.1016/j.annonc.2026.04.008>). These data suggest that thrombocytopenia associated with MCC-DM1 ADCs is likely driven by nonspecific ADC uptake and intracellular accumulation of nonpermeable DM1-containing metabolites rather than by premature systemic release of the DM1 payload. In contrast, uptake of SPP-DM1 ADCs involves internalization of more extensively deconjugated ADC species¹⁴ and the release of free DM1 (membrane-permeable), which may reduce intracellular accumulation of active metabolites. This further illustrates the complexity of ADC pharmacology, in which both linker stability and the properties of the released metabolite influence the clinical profile. Recent

clinical data for XMT-1536 and XMT-1592, which share the same antibody and payload but different conjugation, linker, and DAR, revealed substantial differences in their PK, MTD, and toxicity profiles.^{114,115} XMT-1592 (more stable and homogeneous, DAR ~6) showed higher incidence of pulmonary toxicities than XMT-1536 (less stable and heterogeneous, DAR ~10-12).¹¹⁵ Additional comparisons between similar ADCs with different linker stability, especially among emerging TOPO1i ADCs, remain limited and are complicated by differences in payload structure, which can independently influence ADC potency, bystander effect, PK, toxicity, and efficacy. Trastuzumab brengitecan, for example, is a stable trastuzumab ADC employing a TOPO1i payload (Ed-04), DAR8, and with GGFG linker,¹¹⁶ which could be benchmarked against T-DXd. However, Ed-04 contains one additional methyl group compared with DXd,¹¹⁷ which may contribute to the observed higher hematological toxicities of trastuzumab brengitecan.¹¹⁸

As discussed, although increased ADC linker stability can enhance conjugated payload exposure, it does not necessarily reduce free drug exposure.¹⁰⁵ As a result, stabilizing the linker may shift the toxicity profile, potentially increasing toxicities related to ADC uptake without meaningfully decreasing free payload-mediated toxicities. Changes in ADC disposition by altering linker stability can still cause expected toxicities and lead to the emergence of unexpected ones. This is consistent with ADC clearance occurring primarily via cellular uptake and catabolism, which is largely target-independent and off-tumor.⁷⁷ It is important to note that increased ADC linker stability has, at times, been associated with lower rates or severity of certain toxicities. A clear example is the decreased incidence of neuropathy and neutropenia observed with certain more stable MMAE ADCs, including MYTX-011, Ila-V, and DMUC4064A.^{80,82,83} Despite improvements, these and other more stable MMAE ADCs have been discontinued as dose-dependent ocular toxicities may have emerged before severe neuropathy or neutropenia. The overall balance of safety, efficacy, and patient QoL must be evaluated on a case-by-case basis for each ADC, rather than assuming that improved stability automatically translates into safer and more effective clinical outcomes.

Increased linker stability does not necessarily improve clinical responses

In clinical settings, response rates have typically been evaluated at one dose level identified from dose escalation, and more recently at multiple dose levels as recommended by the FDA Project Optimus. An ADC with a more stable linker may confer improved efficacy but also an unacceptable toxicity profile, or it may not provide greater efficacy compared with a less-stable construct. This is because efficacy reflects a delicate balance of multiple mechanisms, with safety representing an integral component. If unexpected life-impairing or life-threatening toxicities (e.g. ocular toxicity, bleeding, pulmonary toxicity) arise at lower-than-efficacious doses with

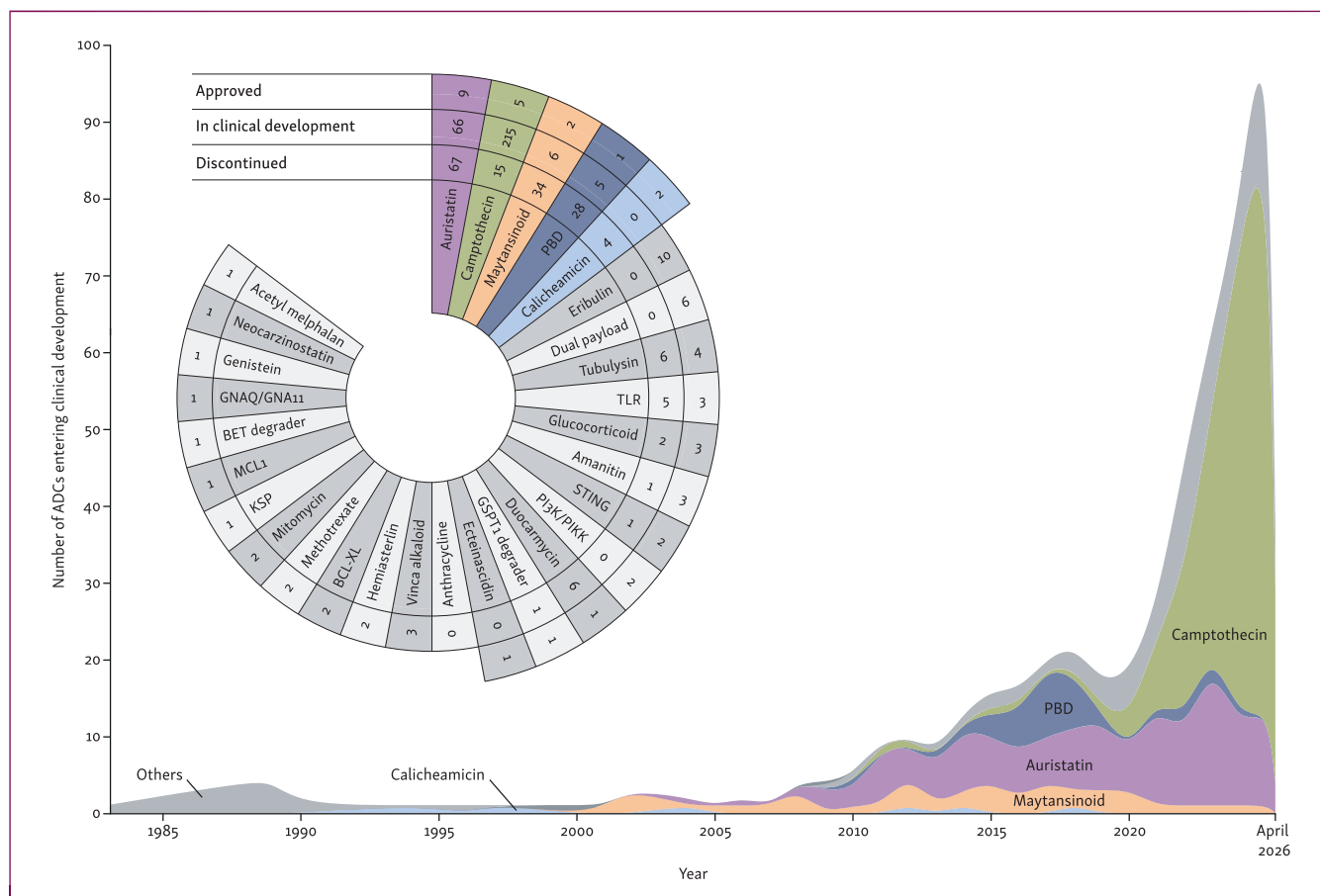


Figure 5. Current ADC landscape as of March 2026. More than 300 ADCs are in clinical development. Payload class is undisclosed for 17 ADCs. The landscape encompasses 557 ADCs that have entered the clinic, 19 approved ADCs (14 approved by US Food and Drug Administration with additional 5 approved only by China NMPA), 342 ADCs in clinical development and 196 discontinued ADCs. The figure illustrates a rapid expansion of camptothecin-based ADCs over the past 5 years, alongside the emergence of new payload classes and dual-payload ADCs. Updated from reference.³ ADC, antibody–drug conjugate; NMPA, National Medical Products Administration.

more stable linkers, the overall efficacy may not surpass that of a similar ADC with a less-stable linker.

Separate trials evaluated two CD79b-targeting ADCs that shared the same antibody paratopes and vedotin drug-linker but differed in linker stability, in comparable patient populations. Although efficacy and response rates were similar, the more-stable ADC (Ila-V) was associated with ocular toxicity, and no ocular toxicity was reported for the less-stable ADC Pola-V (Table 2, Supplementary Figure S8, available at <https://doi.org/10.1016/j.annonc.2026.04.008>). Ultimately, Ila-V was discontinued and only Pola-V received regulatory approval. In the case of MUC16-targeting vedotin ADCs, the more-stable DMUC4064A showed potentially better efficacy compared with the less-stable sofituzumab vedotin. However, DMUC4064A was associated with a high incidence of ocular toxicity, and both ADCs were ultimately discontinued.^{81,82}

For TOPO1i ADCs fewer comparisons exist thus far. T-DXd and GQ1005 both utilize the same trastuzumab antibody, GGFG-AM cleavable peptide self-immolative linker, and DXd payload but feature different antibody-linker stability and DAR (Supplementary Table S2 and Supplementary Figure S12, available at <https://doi.org/10.1016/j.annonc.2026.04.008>).

Although data for GQ1005 remain immature relative to T-DXd, cross-trial comparisons indicate similar toxicity profiles, but lower objective response rates and less deep responses for the more-stable GQ1005 compared with T-DXd in patients with HER2-positive gastric and breast cancers, and HER2-low breast cancer.^{51,119,120}

Beyond these head-to-head examples, several ADCs specifically designed to improve upon prior technologies through more stable linkers have been discontinued from clinical development. Notable recent terminations include MYTX-011 (which employed thiomab site-specific conjugation to increase antibody-linker stability)¹²¹ and EO-3021 (which employed enzyme-mediated site-specific conjugation at glutamine Q295, resulting in a stable ADC).¹²² A differentiated linker, even with an identical antibody and payload, may not necessarily offer improvements over a less-stable predecessor, despite promising preclinical data. SGN-35T retained the same antibody and payload as the approved brentuximab vedotin (BV) but incorporated a novel linker (MP-dLAE) designed to preferentially release MMAE in cancer cells rather than in bone marrow.¹²³ However, SGN-35T was recently discontinued (NCT06120504), highlighting its apparent lack of superiority

over BV. TOPO1i ADCs are relatively recent and have experienced fewer program terminations than other payload classes (Figure 5).³ Nonetheless, reported discontinuations among TOPO1 ADCs (Supplementary Table S4, available at <https://doi.org/10.1016/j.annonc.2026.04.008>) span the full spectrum of linker stability, including constructs with low stability (IMMU-130), moderate stability (MC-based DS-6157, GEN1160, and GEN1286), high stability (PF-08046044, AZD9829, PHN-010, BAT8007, and BAT8009), and fully stable linkers (PF-08052666 and 9MW2921).

Finally, with industry focus shifting toward the use of new payload mechanisms to overcome resistance, dual-payload ADCs have started to enter clinical development (Figure 5). Incorporating two or more distinct payloads further complicates linker design and the determination of optimal stability. The resulting complexity requires careful PK assay design and may render the attribution of specific toxicities to individual payloads or to linker instability impractical.

CONCLUSION AND OUTLOOK

This article highlights the complexity of identifying ADCs with optimal linker stability across different payload classes, the pitfalls of methods used to evaluate linker stability, and limitations of preclinical species in predicting clinical payload exposure. Linker instability is often portrayed as inherently negative, but this view is not necessarily supported by data. Indeed, a defined degree of linker instability may contribute to favorable ADC pharmacology by balancing payload release, systemic exposure, and clearance pathways.

Achieving the right balance in ADC design is complicated by the multivariable and dynamic nature of ADCs. Free payload exposure is dependent on linker stabilities, ADC clearance, and the physicochemical and metabolic properties of the payload itself, which dictate how it is cleared. Thus, it is unsurprising that the impacts of linker stabilization appear payload-dependent and optimal stability may vary across payload classes. ADCs with high rates of linker-drug instability in circulation (half-lives of 1–2 days), including SG and Sac-TMT, are clinically approved. This highlights that effective ADC designs can also incorporate linker instabilities and extracellular payload release rather than relying solely on ADC cellular uptake.

A preoccupation with linker stability overlooks ADC disposition: (i) ADCs are not eliminated from the body without being processed; therefore, all payload will be released via a combination of cellular uptake and catabolism, extracellular linker cleavage, and linker instabilities; (ii) the majority of ADC-injected dose is catabolized in normal tissues; (iii) stabilizing ADCs often improves antibody-conjugated drug exposure but may shift rather than prevent payload release, potentially leading to unexpected toxicities; (iv) ADC tumor uptake and systemic payload exposure (resulting from multiple mechanisms, including ADC catabolism and linker instabilities) ultimately

contribute to tumor payload exposure and, consequently, ADC efficacy; (v) ADC uptake in normal tissues and systemic payload exposure contribute to toxicities.

These considerations suggest that improving clinical outcomes for ADCs may depend more on optimizing payload properties and designing new payloads, rather than on increasing linker stability alone. Because of significant interspecies differences, and despite allometric scaling, the impact of conjugated drug and free payload exposure may ultimately be understood only in the clinical setting. New preclinical strategies to estimate payload effects could be valuable, such as slow dynamic infusion of payload, although their ability to fully recapitulate clinical outcomes is unproven. Novel linker designs remain important tools to fine-tune ADC properties, including overall clearance and the rate and site of payload release. However, in contrast to preclinical findings which clearly highlight benefits of linker stabilization, clinical data suggest that focusing solely on increasing linker stability does not guarantee better ADCs. Instead, achieving the right degree of stability is essential to balance efficacy and toxicity. Nuances in ADC structure can have a profound impact on patient outcomes, underscoring the importance of refining our understanding based on clinical evidence rather than preclinical promise. Truly tumor-selective cleavable linkers could represent a paradigm shift in ADC design by preventing payload release in normal tissues, although such linkers have not yet been realized.

Analysis of clinical PK, particularly when comparing constructs with the same payload and antibody but different linkers, can yield insights into the impact of linker modifications. Additionally, understanding the strengths and limitations of PK and linker stability assays is important for cautious data interpretation (Table 1). If the number of analytes are limited, measuring acDrug in combination with TAb and free payload is more informative than total ADC.¹²⁴ PK data should be reported in nmol/L (nM) to facilitate comparisons, given significant differences in molecular weight between antibodies and payloads. Clinical quantification of albumin–drug conjugates could broaden our understanding of ADCs that undergo thiol–maleimide exchange. Finally, because payloads reach pharmacologically active and toxic levels in humans, understanding their properties, including clearance and metabolism, is critical. For instance, SN-38 undergoes glucuronidation, a metabolic pathway not shared by most TOPO1 inhibitors, resulting in characteristic gastrointestinal toxicities.

With increasing focus on next-generation payloads, the lessons learned from decades of ADC clinical experience should help guide their rational design by tailoring payload properties to ADC modalities. Overall ADC disposition, stability considerations, and translational limitations highlighted in this article will likely apply to ADCs with different antibody formats (e.g. biparatopic, bispecific), novel payloads, and dual payloads. However, novel constructs will bring new challenges that current frameworks and assumptions may not fully anticipate. Many new ADCs (Figure 5) with innovative linker technologies are in clinical

evaluation. These studies may help clarify how linker stability contributes to therapeutic index, safety, and anti-tumor activity across different ADCs and payload classes. Ultimately, we hope that a better understanding of these relationships can be leveraged to improve patient outcomes.

ACKNOWLEDGEMENTS

We would like to acknowledge Patricia M. LoRusso, Barbara Pistilli, Paul A. Moore, and Shang-Chiung (Charles) Chen for their valuable insights and feedback, which greatly contributed to the development of this article.

FUNDING

None declared.

DISCLOSURE

RC reports a relationship with Zymeworks BC Inc that includes employment and equity or stocks; a relationship with AstraZeneca PLC that includes equity or stocks. SDS reports a relationship with Zymeworks BC Inc that includes employment and equity or stocks. SDB reports a relationship with Zymeworks BC Inc that includes employment and equity or stocks. JRR reports a relationship with Zymeworks BC Inc that includes employment and equity or stocks.

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